

35. .Implement document storage and top-k retrieval using a vector database.

Code:

```
from sentence_transformers import SentenceTransformer
import faiss
import numpy as np

# Load embedding model
model = SentenceTransformer("all-MiniLM-L6-v2")

# Documents to store
documents = [
    "Python is a high level programming language.",
    "Machine learning is a method of learning from data.",
    "Artificial intelligence creates intelligent computer systems.",
    "Natural language processing helps computers understand human language.",
    "Python is commonly used for data science.",
    "Deep learning uses artificial neural networks.",
    "Computer networks connect computers and devices.",
    "Database management systems store and manage data."
]

# Convert documents into embeddings
document_embeddings = model.encode(documents)

# Convert embeddings to NumPy float32
document_embeddings = np.array(
    document_embeddings
).astype("float32")
```

```
# Get vector dimension
dimension = document_embeddings.shape[1]

# Create FAISS vector database
database = faiss.IndexFlatL2(dimension)

# Store document embeddings
database.add(document_embeddings)

print("DOCUMENT VECTOR DATABASE")
print("-----")
print("Total documents stored:", database.ntotal)

# Get query from user
query = input("\nEnter your query: ")

# Convert query into embedding
query_embedding = model.encode([query])
query_embedding = np.array(
    query_embedding
).astype("float32")

# Ask user for top-k value
k = int(input("Enter number of documents to retrieve (K): "))

# Search the vector database
distances, indices = database.search(
    query_embedding,
```

```

        k
    )

# Display retrieved documents
print("\nTOP-K RETRIEVED DOCUMENTS")
print("-----")

for i in range(k):
    document_number = indices[0][i]

    print("\nRank:", i + 1)
    print("Document:", documents[document_number])
    print("Distance:", round(
        float(distances[0][i]), 4
    ))

print("\nRetrieval completed successfully.")

```

output:

```
ff.py - C:/Users/Dhanyashree/ff.py (3.10.11)
File Edit Format Run Options Window Help
# Convert embeddings to Numpy float32
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print("\nRetrieval completed successfully.")

IDLE Shell 3.10.11
File Edit Shell Debug Options Window Help
Python 3.10.11 (tags/v3.10.11:7d4cc5a, Apr 5 2023, 00:38:17) [MSC v.1929 64 bit
(AMD64)] on win32
Type "help", "copyright", "credits" or "license()" for more information.
>>>
===== RESTART: C:/Users/Dhanyashree/ff.py =====
DOCUMENT VECTOR DATABASE
-----
Total documents stored: 8

Enter your query: Python for data science
Enter number of documents to retrieve (K): 3

TOP-K RETRIEVED DOCUMENTS
-----

Rank: 1
Document: Python is commonly used for data science.
Distance: 0.2768

Rank: 2
Document: Python is a high level programming language.
Distance: 0.719

Rank: 3
Document: Database management systems store and manage data.
Distance: 1.3548

Retrieval completed successfully.
>>>
```